

1	(a)	(i)	Archimedes' Principle states that the <u>upthrust</u> on a body completely or partially submerged in a fluid is <u>equal</u> in magnitude and opposite in direction <u>to the weight of the fluid</u> the body <u>displaces</u> .	B1
		(ii)	Pressure increases with depth in a fluid. When an object is submerged in a fluid, bottom is at a greater depth than at the top, hence, the <u>pressure is greater at the bottom than at the top of the object</u> . The difference in pressure <u>results in a net upward force</u> acting on the object, which is upthrust.	B1 B1
	(b)		$F_{\text{net}} = 0$ $m_{\text{total}}g = U + F_{\text{resistive}}$ $m_{\text{submarine}}g + m_{\text{seawater}}g = \rho g V_{\text{submarine}} + F_{\text{resistive}}$ $4800(9.81) + m_{\text{seawater}}(9.81) = (1030 \times 9.81 \times 5) + 1100$ $m_{\text{seawater}} = 462 \text{ kg}$	M1 A1
	(c)	(i)	$P = P_o + \rho gh$ $= 1.01 \times 10^5 + (1030 \times 9.81 \times 200)$ $= 2.12 \times 10^6 \text{ Pa}$	M1 A1
		(ii)	$\Delta P = P_{\text{outside}} - P_{\text{inside}}$ $= 2.12 \times 10^6 - 1.01 \times 10^5$ $= 2.02 \times 10^6 \text{ Pa}$ $F = \Delta P A = 2.02 \times 10^6 (\pi \times 0.150^2) = 1.43 \times 10^5 \text{ N}$	M1 A1

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2	(a)	Gravitational potential at a point is the work done by an external force per	R2
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